

Justification for Other than Full and Open Competition
at or Below the Simplified Acquisition Threshold

Project Title: Systems biology of immune correlates of novel universal vaccine development and testing

I. Identification of the agency and contracting activity

DHHS, NIH, NIAID, AMOB-BETHESDA (AMOB)

Requestors name:

Requestors phone

Requestors email

Address of the req

II. Description of the project, and supplies or services required to meet the agency's needs (including the estimated value)

This is a follow-on purchase of a bioinformatics analysis program currently used by the VPES to perform comprehensive statistical modeling of 'multi-omics' data and advanced visualizations for target-of-interest, biological network characterization, and diagnostic/prognostic biomarker identification in genomics/transcriptomics/proteomics data sets from animal model and human clinical samples derived from vaccine and infection studies performed by the VPES, as part of its mission to develop novel universal influenza vaccines.

A. Type of action. Check the appropriate box:

☐ New requirement

☒ Follow-on

- Name of the previous contractor(s): Genedata
- Sequence number(s), if available: 2253271
- Was the initial and any subsequent awards competed? ☐ Yes ☐ No

B. AMBIS Sequence number for this requirement: 2277910

C. Total estimated dollar value of the requirement (including options, if applicable): \$ 25,900

D. Period of performance, *inclusive of options*: from Feb 1, 2026 to Jan 31, 2027

E. Project Background

The **Viral Pathogenesis and Evolution Section** (VPES) within Laboratory of Infectious Diseases (LID) conducts research directed toward defining the pathobiology of respiratory viral pneumonia, including SARS-CoV-2 and influenza A viral infections and developing means for their control. VPES engages in a wide range of research activities that extends from clinical virology and correlates of immunity studies in humans to molecular characterization of animal and human infections with influenza A viruses that can cause severe, acute disease of the respiratory tracts in animal models. Scientists in the section employ high-throughput genomic, proteomic, computational modeling, and machine learning to understand the molecular basis of viral disease and the pathogenesis viral pneumonia in humans and animals and with the goal to develop and evaluate novel vaccines, antiviral drugs, and therapies. VPES has developed a novel universal betacoronavirus vaccine that is in pre-clinical development and also a novel universal influenza vaccine that is in human clinical trials.

F. Project Description

To perform bioinformatics and statistical analysis of large 'multi-omics' data sets to characterize the host gene and

protein expression responses in human and animal model samples during universal betacoronavirus and universal influenza vaccination and/or infection with influenza A viruses.

III. **Identification of the statutory authority permitting other than full and open competition (SELECT ONE)**

- ☐ **41 USC 253(c)(1) and FAR Part 6.302-1: Only one responsible source**
- ☒ **41 USC 501 and FAR Part 8.405-6: Federal Supply Schedule (FSS), limited competition**
 - ☐ An urgent and compelling need exists, and following the procedures would result in unacceptable delays
 - ☒ Only one source is capable of providing the supplies or services required at the level of quality required because the supplies or services are unique or highly specialized; or
 - ☐ In the interest of economy and efficiency, the new work is a logical follow-on to an original Federal Supply Schedule order provided that the original order was placed in accordance with the applicable Federal Supply Schedule ordering procedures. The original order or BPA must not have been previously issued under sole-source or limited-sources procedures.

***Important Note:** The statutory authorities listed below will rarely ever be used in NIH. Before selecting one of these authorities, please ensure that you have read and understood the citation and discussed it with the cognizant Contracting Officer.*

- ☐ **41 USC 253(c)(2) and FAR Part 6.302-2: Unusual & compelling urgency**
- ☐ **41 USC 253(c)(3) and FAR Part 6.302-3: Industrial mobilization; or engineering, developmental, or research capability; or expert services**
- ☐ **41 USC 253(c)(4) and FAR Part 6.302-4: International agreement**
- ☐ **41 USC 253(c)(5) and FAR Part 6.302-5: Authorized or required by statute**
- ☐ **41 USC 253(c)(6) and FAR Part 6.302-6: National security**
- ☐ **41 USC 253(c)(7) and FAR Part 6.302-7: Public interest**

IV. **Demonstration that the proposed contractor(s) unique qualifications or the nature of the acquisition requires use of the authority cited:**

A. Name and address of the proposed contractor(s)

Genedata (USA), Inc.
750 Marrett Road
One Cranberry Hill, Suite 403
Lexington, MA 02421

B. Describe why only the requested supply/service will meet the government's needs

This is a follow-on purchase Genedata Analyst required for project continuity. Genedata Analyst utilizes rigorous data normalization, statistical algorithms, machine learning and interactive data analysis and visualization tools. Genedata Analyst also includes proprietary mapping features for integration of diverse, complex data from both public domain data with proprietary experiments and combination of genomics profiling and phenotype data correlation. Specifically, Genedata Analyst provides the following features. (1) Data normalization tools, including linear normalization to standardize experiments and remove artifacts and nonlinear normalization methods like LOWESS, Quantile Normalization and Median Polish. (2) Statistical tools using both parametric and non-parametric tests, including t-Test, mixed linear models (ANOVA and ANCOVA), Principal Components Analysis (PCA), multiple testing corrections controlling false positives from high-throughput experiment and trend identification and time series analyses that support complex clinical metadata-based analysis. (3) Machine learning analysis tools including modeling and prediction Decision Trees, Partial Least Square analysis (PLS), Linear Discriminant Analysis, feature selection methods like ANOVA and Recursive Feature Elimination and model validation using leave-one-out and Monte Carlo cross validation. (4) Interactive data visualization, including histograms, bar charts, box plots, heat maps, scatter plots, trees, maps, and parallel coordinate plots, and proprietary specialized viewers for understanding the results of complex statistical analyses and biological interpretation. All data visualizations in Analyst are fully interactive and feature shared selections between all graphs to facilitate and support interactive data exploration.

C. Describe why only the proposed contractor is capable of providing the requested supply/service

Integration of a diverse set of data normalization, statistical, data visualization, and proprietary data mapping tools are critical VPES research on influenza viral diseases and completion of its mission. Moreover, VPES has in excess of 100,000,000 gene expression data points from over 8 years of clinical and experimental vaccine development studies, that are critical for on-going long-term, clinical immunology and universal vaccine testing studies.

V. **Certification**

This is to certify that the justification for the proposed acquisition has been reviewed and that to the best of my knowledge and belief the information and/or data provided to support my request (for the requestor) and approval (for the CO) of is accurate and complete.

Official	Name & Title	Signature	Date
Requestor Certification			
Contracting Officer Approval			